

AV System (Technical Specs)

SL	Specification	Qnt.	Unit.
1.	<u>Supply, installation, & Commissioning of the Chairman Unit Long Length - Digital</u> Technical specifications Frequency response:200 Hz to 12.5 kHz Headphones load impedance:> 32 ohms < 1k ohm Loudspeaker nominal output:72 dB SPL Length of microphone :400 mm or more Operating temperature:0 °C to +35 °C Plug and play functionality for ensuring fast installation and reconfiguration times Immunity to mobile phone interference Functions Possible-to-speak indication Request-to-speak indication Built-in loudspeaker with volume control Built-in 3.5 mm (0.14 in) stereo headphones socket Built-in volume control for headphones Colour-coded LED indicator above microphone button for showing the following states: microphone on, request-to-speak, and possible-to-speak	1	Nos.
2.	<u>Supply, installation & Commissioning of Delegate Unit Long Unit-Digital</u> Technical specifications Frequency response:200 Hz to 12.5 kHz Headphones load impedance:> 32 ohms < 1k ohm Loudspeaker nominal output:72 dB SPL Length of microphone :400 mm or more Operating temperature:0 °C to +35 °C Plug and play functionality for ensuring fast installation and reconfiguration times Immunity to mobile phone interference Functions Possible-to-speak indication Request-to-speak indication Built-in loudspeaker with volume control Built-in 3.5 mm (0.14 in) stereo headphones socket Built-in volume control for headphones Color-coded LED indicator above microphone button for showing the following states: microphone on, request-to-speak, and possible-to-speak	15	Nos.
3.	<u>Supply, installation & Commissioning of Digital Control Unit</u> <u>Built-in MP3 recorder with internal memory and USB recording</u> Built-in Digital Acoustic Feedback Suppression Quick and easy connection upto 80 discussion devices Native support for automatic HD camera control Signal to noise :> 94 dBA Frequency response for input :30 Hz to 20 kHz Recording/ Playback (bit rate):64, 96, 128, 256 Kbit/sec Nominal loudspeaker output:72 dB SPL Operating temperature: 5 °C to +45 °C Relative humidity: 5% to 98% non-condensing Functions Plug-and-play functionality Web browser control Open microphone control Built-in MP3 recorder Discussion control	1	Nos.

4.	Supply, installation & Commissioning of EXTENSION CABLE (8 pin: cable should be made of 100% pure copper) or CAT 6 cable as per site requirement (for connecting 1st and the last delegate to the controller to form a ring topology).	1	Lot
5.	Supply, installation & Commissioning of 30-w wall-mount loudspeaker Maximum Power: 45 W, Rated Power: 30 W, Power Tapping: 30/15/7.5 W, Sound pressure level at rated power / 1 W (1 kHz, 1 m): 105 / 90 dB (SPL), Effective frequency range (-10 dB): 100 Hz to 18.5 kHz, Opening angle at 1 kHz / 4 kHz (-6 dB): 160° / 81° (hor.) 150° / 90° (ver.), Transducers: 127 mm (5") woofer, Weatherproof IP65 protection, Safety according to IS 616:2017 & IS 15596-5, Operating Temperature: -25 °C to +55 °C	4	Nos.
6.	Supply, installation & Commissioning of 240-WATT MIXER AMPLIFIER with Bluetooth and USB 4 microphone inputs with microphone/line Inbuilt Bluetooth for audio streaming; Supports audio playback from USB pen drive S/N at rated power: more than 60 dB (mic); >65 dB (line) Frequency response: 50Hz to 20kHz Rated Power :240W Operating temperature: -10 °C to +55 °C	1	Nos.
7.	Digital Signal processor (DSP): The DSP shall have inbuilt Acoustic Echo Cancellation The DSP shall be a network-enabled conferencing processor with range of 8–16 channels of AEC, 8–12 mic/line inputs with phantom power, 4–8 outputs, USB audio interface, and advanced DSP including automixing, EQ, AGC (Auto Gain Control), Noise reduction (ANC), and feedback suppression. The system shall support 32/40-bit floating-point processing with a frequency response of 20 Hz–20 kHz and THD <0.005%. The processor shall support Ethernet control, scalable architecture, and integration with UC platforms like Teams/Zoom. BIS certified	1	Nos.
8.	Supply, installation & Commissioning of Handheld Wireless Microphone. UHF dual handheld Wireless Mic system set with two transmitters dynamic microphone, Frequency Response 40Hz-15,000 Hz, Surge mode PLL Phase lock, Dynamic Range >90dB, THD < 0.5%, S/N ratio > 90dB, Power Consumption < 85mA.	1	Nos.
9.	Supply, installation & Commissioning of Dual Lapel Wireless Microphone UHF Lapel Wireless Mic system set with two transmitters dynamic microphone, Frequency Response 40Hz-15,000 Hz, Surge mode PLL Phase lock, Dynamic Range >90dB, THD < 0.5%, S/N ratio > 90dB, Power Consumption < 85mA. -	1	
10.	Supply, installation & Commissioning of Cameras for Tracking purpose with look at me features 20X PTZ Camera with 4 K resolution - PTZ camera shall have 1/2.8" CMOS 8.51 sensor or better ; S/N Ratio of >55db ; It shall have 20x optical zoom or better ; along with Backlight compensation ; shall have USB 3.0 port for unified communication applications, HDMI/DVI Port for Hardware Equipment & IP interface for monitoring; Network protocol : TCP/IP, HTTP, RTSP, RTMP/RTMPS, SRT, Onvif, DHCP, Multicast ; Video stream :It shall support Dual H.264 and H.265 Streams ; It shall support upto 60 fps or better. Support UVC and UAC. It shall have 60° or better field of view; with atleast 200 presets; Pan rotation: Pan Rotation Angle of -170degrees to +170 degrees or better ; Tilt Rotation Angle of -30 degrees to + 90 degrees or better. It shall support Horizontal and Vertical Flip ; Certification: BIS, ISO 9001:2015 Function: One camera should focus on the chairman and other should be able to track.	2	Nos.

11.	Speaker Tracking System + 4 port Splitter -4K HDMI to USB 3.0 Video Capture Device,4K HDMI 2.0 to USB Type-C Video Capture Resolution up to 4K@30fps and 1080P@120fps 3.5mm Mic-in Mixed with HDMI Audio and 3.5mm Audio Line-out Supports 4K HDMI Loop- through and HDMI to USB-C Capture along with 4-Port 4K HDMI Splitter, Connects one HDMI source to four HDMI displays at the same time HDMI (3D, Deep Color, 4K); HDCP Compatible Long distance transmission – up to 15 m (24 AWG) Supports max. data rate of 10.2 Gbps Additional HDMI matrix switcher with audio de embedder: 4 Input 4 Output HDMI Switch, 4K Ultra HD HDMI Switcher Support 3D, 1080P, 4Kx2k@60Hz (IR Remote Control)	1	Nos.
12.	Supply, installation & Commissioning of 24 port manage switch or better as per the requirement	1	Nos.
13.	Supply, installation & Commissioning of Interactive Display 98 octa core Processor , Android 14 or above, Touch Point 40 of reputed brand: with inbuilt apps & smart TV display features. IS 13252 (Part 1): 2010 & IS 616 : 2017 certified. HDMI cable from connectivity to source	1	Nos.
14.	Supply, installation & Commissioning of 42/43-inch LED display Android OS, 4GB RAM, 32GB Storage of reputed brand with WIFI connectivity (to be mounted on portable stand) HDMI cable for connectivity – 5 meters X 2 units. IS 13252 (Part 1): 2010 & IS 616 : 2017 certified.	2	Nos.
15.	Portable TV Stand ; with wheel base to move it around freely; Strong heavy-duty model to make sure your LED TV lies in safe hands with no risk of falling breaking at the same remaining portable; Load capacity: 25 Kgs;	2	Nos.
16.	Supply, installation & Commissioning of 55 inch LED display Android OS, 4GB RAM, 32GB Storage of reputed brand with integrated connection to AV system. IS 13252 (Part 1): 2010 & IS 616 : 2017 certified.	1	Nos.
17.	Supply, installation & Commissioning of Podium Gooseneck Microphone having LED status display and with programmable button for Push to Talk or Push to Mute functions	1	Nos.
18.	Supply, installation & Commissioning of Acrylic podium with color CCRH Logo which can be easily movable suitable for 5-6 feet height persons	1	Nos.
19.	Supply and installation of TAB of reputed brand for Chair person 1 & Chair person 2 - with Android 14 or higher operating system with stylus; Screen size: 12 inches or Higher with AI Writing, Desktop Mode, Pre Loaded Pro Apps; works only on Wifi; with minimum 12 GB RAM, 256 GB storage; resolution : 2360*1640p or better; primary camera : minimum 12 MP; Multi-Touch display with IPS technology	2	Nos.
20.	Supply, installation and commissioning of wireless presentation - through which all the presentations can be shared with the interactive screen through laptops ; it should also have the provision for the chairman to give this suggestions/ remarks / corrections through the TAB ; and the same should be visible in the main screen Features :- 4K - Wireless Screen Sharing - Support for AirPlay, Miracast, and Chromecast for app-free, dongle-free screen mirroring from any device ; BYOM - Bring Your Own Meeting Web Conferencing : Host professional web meetings wirelessly from a laptop -using camera, speakers and microphones already installed in the room. Enterprise Level Security - Non-modifiable Linux system with wireless data encryption, user access management, and a range of customizable security features Additional features- Flexible Network Integration, Whiteboard & Annotation, Digital Signage Player, Room Management APIs Operating system : Linux; Output Res : 2160p60/30, 1080p60/30, 720p60; Supported devices: iOS, iPadOS, Android, Windows, macOS, Chrome OS; Supported mirroring	1	Lot

	protocols: AirPlay, Miracast (Peer-to-Peer and MICE), Chromecast, vSolution Cast; Max No of screen windows :4		
21.	Supply , installation of Laptop of reputed brand (i7, 14th generation, 32 GB RAM, 3200 GHZ, 1 TB SSD: Additional wireless mouse and wireless keyboard, Windows 11 Professional)	1	Nos.
22.	Supply, installation of electrical work and carpentry work as required with accessories (like switch, socket, cable manager installation for conference room participants). For this the bidders may visit the institute during the bid time.	1	Lot
23.	Supply, installation of access point including cable work from main connection to the conference WiFi 6, 2 x 2 MU-MIMO, Dual-Band, Multi-Gig PoE+, Wireless Access Point with , 802.3af/at PoE/PoE+, 1.8 Gbps throughput with 1 year cloud subscription included Should Support 5G Prior for a faster Ethernet. Wireless Security: Open, WPA, WPA2PSK_TKIPAES, WAP2_EAP, 802.1x,	1	Lot
24.	Supply, installation & Commissioning of all Speaker cable -2 core 1.5 sq mm unarmoured with casing; all HDMI Cables	1	Lot
25.	Supply and installation of Hydraulic Cable Cubby Should have HDMI, Audio, LAN and Power Ports of reputed make Designed for perfect integration into the office furniture. Soft opening with comfort and safety. Connectivity: 2 x Universal Power 2 x USB-A Mobile Charger, 1 x HDMI 2.0V, 1 x Type-C USB 3.1V, 2 x CAT 6A Network/RJ-45. Colour should be suitable and apt for the furniture already installed in the conference room.	5	units
26.	Supply of installation 3 KVA online UPS with 30 minutes backup for the complete system of reputed make	1	unit
27.	Supply, installation of 24U Fully Loaded Floor Standing Network Rack (600x800) with accessories or better	1	Lot
28.	3 years of Warranty and 3 years of CMC should be provided		